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Motivation

We want to know whether we can predict power for generative AI jobs from signals we can actually observe: training vs. inference, LLM vs. image generation, node count, and related runtime features. That is a planning problem, not just a modeling exercise. Power and cooling get sized before the job mix is known.

Most published server-energy models were built on older cloud work. Lin et al. (2020) and Chuang et al. (2023) get under 1 W average error on CPU-, memory-, I/O-, and mixed jobs. An H100 floor does not look like that stack. Chauhan (2024) puts GPU/TPU-heavy AI compute at about 3-to-5 times the power of the older mix. If that gap is real in measured traces, a model fit only on classical workloads will miss the load that now matters.

We use NLR’s generative AI power profiles (Vercellino et al., 2026) for that reason. The goal is energy forecasts for current AI training and inference, not another model of yesterday’s servers.

Data:

Our main source of data comes from a dataset of generative AI workload power profiles from National Laboratory of the Rockies (NLR). The dataset captures time-resolved electrical power measurements in high-performance computing data center equipped with NVIDIA H100 GPUs, including variations in job type (inference vs. training), workload (LLM vs. image generation), datasets, and number of compute nodes. The data link is <https://data.nlr.gov/submissions/312>
Reference: Vercellino et al., “*Measurement of Generative AI Workload Power Profiles for Whole-Facility Data Center Infrastructure Planning*,” [arXiv:2604.07345](https://arxiv.org/abs/2604.07345) (2026).

We also look at a Kaggle dataset measuring the energy-efficient AI system performance. This dataset contains 7167 records capturing system-level energy usage, computational performance, and resource utilization metrics across diverse operating conditions. It is designed to support analysis of energy efficiency, system behavior, and performance trade-offs in AI-driven environments. The data link is:

<https://www.kaggle.com/datasets/colabssss/energy-efficient-ai-system-performance-dataset>

Related Work:

There is a growing body of research aimed at predicting data center energy consumption using machine learning. Various nonlinear modeling approaches have been explored, such as gradient boosting and multilayer neural networks, which aim to model energy consumption

using server workload and hardware features. For example, Lin et al. proposed two neural network models which predict energy consumption across four data center workloads: CPU-intensive, memory-intensive, I/O-intensive, and mixed workloads (2020). Using features that track memory usage, I/O activity, and processor times, they were able to predict energy consumption across the four workloads with less than 1 Watt average prediction error. Chuang et al. achieved similar accuracy (less than 1 Watt average error) using distance correlation feature extraction and gradient boosting across the same four workloads (2023).

We aim to build on this body of research by creating a model oriented around modern, AI-focused workloads, rather than traditional cloud architecture and job types. AI workloads have unique hardware and power requirements, requiring 3-to-5 times more power for complex TPU and GPU-bound compute (Chauhan, 2024). Energy consumption models that emphasize classical hardware stacks and compute patterns may not extend to power-intensive AI server activities and infrastructure. In order to capture the hardware and energy profiles of these workloads, our model will leverage the high-resolution features of the NLR dataset, such as job types, model architectures, prompt frequency, and output token lengths, generating energy forecasts tailored specifically to AI training and inference.

References

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